

Meeting 2

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Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
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Outcome of previous meeting

Previous meeting, we talked about moving towards coding. For this, we discussed to:

- Narrowing down the set of frameworks based on 'popularity'.
- Narrowing down the set of algorithms.
- Narrowing down the set of environments
- Start with 2 frameworks (and move up to 3), to experiment on an environment with the frameworks own hyperparameters and on the same hyperparameter set.

Results during the last weeks

- Determined which frameworks I will start experimenting with based on impact within literature and popularity amongst researchers.
- Narrowed down the set of potential algorithms for relative ranking across different frameworks, based on overlap in implementation.
- Looked into the intersecting set of environments between SB3 and CleanRL.
- Started building an API that can easily use hyperparameter configurations of other frameworks for structured experimenting and reproduction. The goal is to adjust the original code as little as possible. Adjustments are denoted. Logging of certain parameters is aligned such that the frameworks can be compared. Default hyperparameters are listed in files.
- Ran 2 test experiments on CartPole-v1 and HalfCheetah-v4, but I acknowledge some discrepancy in the experiments.

Plans for next weeks

- Find out how to support Mujoco-v1. Furthermore, find out what Atari version has been used in the original PPO paper. → Experiment to align results of original PPO paper with SB3, CleanRL, and Tianshou using the same hyperparameters to expose consistency between frameworks and original algorithm.
- Similar environment setup across experiments, regarding wrappers (in continuous environments).
- Atari import-model support
- Tianshou as third framework of comparison
- → Run a shared-support set of environments of SB3, CleanRL, and Tianshou on own hyperparameters to find the best configuration. Run on the same environment, PPO of SB3, CleanRL, and Tianshou on the same hyperparameter set to see if implementations are directly comparable with each other